

# VAJRALA GEETHIKA

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## Summary

Aspiring Engineer graduate with a strong foundation in Python, data structures, and object-oriented programming. Skilled in developing efficient, scalable applications using Python with a focus on clean, maintainable code. Experienced in collaborating within the Software Development Lifecycle to debug complex systems and implement test driven development practices. Committed to continuous learning and applying technical problem-solving skills to contribute effectively to software engineering teams.

## Technical Skills

|                                       |                                                                                        |
|---------------------------------------|----------------------------------------------------------------------------------------|
| <b>Programming &amp; Databases:</b>   | Python (Advanced), SQL (MySQL, PostgreSQL)                                             |
| <b>Data Analysis &amp; Stats:</b>     | Exploratory Data Analysis (EDA), Data Cleaning, Statistical Analysis                   |
| <b>Software Engineering:</b>          | Data Structures & Algorithms, Object- Oriented Programming (OOPs)                      |
| <b>Development Lifecycle:</b>         | Software Development Lifecycle (SDLC), Test-Driven Development, Debugging              |
| <b>Machine Learning:</b>              | Supervised/Unsupervised, Scikit-Learn, Model Evaluation                                |
| <b>Reporting &amp; Visualization:</b> | Power BI, Advanced Excel (Pivot Tables, VLOOKUP, Data Validation), Matplotlib, Seaborn |
| <b>Tools &amp; Concepts:</b>          | ETL Pipelines, Git/GitHub, Jupyter Notebook, VS Code                                   |

## PROJECTS

### Loan Default Risk Prediction

Python, SQL, Scikit-learn, Pandas

- Performed extensive data validation and quality checks on 50,000+ financial records to ensure data accuracy before model development.
- Applied advanced normalization and feature engineering techniques, improving model sensitivity by 15%.
- Achieved a 0.82 ROC-AUC score through rigorous hyperparameter tuning and cross-validation, significantly reducing potential credit loss.
- Conducted deep-dive statistical analysis to uncover latent borrower repayment patterns and visualize primary risk factors.
- Investigated data anomalies and exceptions through exploratory analysis, identifying key risk patterns and improving decision-making accuracy.

### Customer Purchase Behavior Analytics

SQL, Python, Power BI

- Analyzed 3,000+ transaction records using Optimized SQL Queries for complex data aggregations and trend discovery.
- Automated routine reporting tasks via Python Scripting, resulting in a 30% reduction in manual data processing time (10 hours/week).
- Architected interactive Power BI dashboards to track critical KPIs and product demand patterns, enhancing stakeholder decision-making speed.
- Improved data reliability for downstream modeling by 15% through systematic data cleaning and outlier detection.
- Conducted quality audits and validation of transactional datasets to improve reporting accuracy and maintain data integrity.

### AI-Powered Computer Vision Attendance System

Python, OpenCV, SQL

- Engineered an automated face-recognition system using OpenCV to manage high volume records, increasing logging accuracy by 25%.
- Maintained accurate records and database consistency for 1,000+ student profiles through regular quality verification processes.
- Performed systematic testing, validation, and exception handling to ensure reliable attendance tracking under varying operating conditions.

## EDUCATION

**SASTRA Deemed University | B. Tech - Electronics and Communication Engineering | Oct 2021 - Sep 2025**

Thanjavur, Tamil Nadu

**CGPA:** 7.5/10

## CERTIFICATIONS

**Python for Data Science:** Microsoft  
**SQL & Database Design with MySQL:** Scaler